Enterobacterales I: General characteristics, Opportunistic members and Associated infections

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Gram Negative Bacilli (GNB)

GNB / Tests	Enterobacterales	Vibrionaceae & Aeromonadaceae	Non-Fermentative Bacteria (NFB)
Oxidase	- *	+	+/-
Glucose	Fermentation	Fermentation	No reaction / Oxidation
TSI	K/A or A/A ± G ± H_2S	K/A or A/A ± G No H_2S	Mostly K/N No Gas
Growth on MacConkey	Growth	Growth	Growth / No growth
MacConkey reaction	LF/NLF	Almost all is NLF	NLF

(* except *Plesiomonas shigelloides*)


เชื้อชนิดหนึ่งเจริญบน BA และ MAC ดังแสดง ถ้าเชื้อนี้ให้ผลการทดสอบ oxidase เป็น negative เชื้อชนิดนี้น่าจะจัดอยู่ในกลุ่มใด?

a. Enterobacterales

b. Vibrionaceae and Aeromonadaceae

c. Non-fermentative bacteria (NFB)

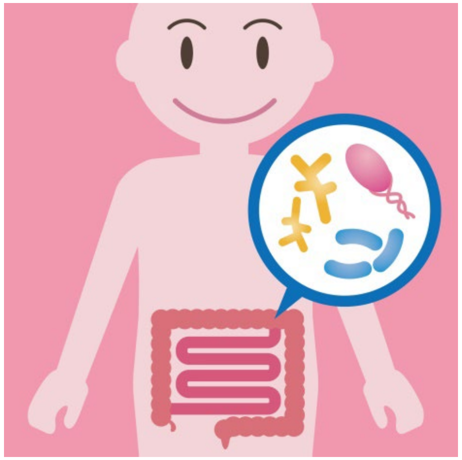
Enterobacterales Overview

- General characteristics
  - Microscopic and colony morphology
  - Classification
  - Virulence and antigenic factors
  - Clinical significance

- Opportunistic members and associated infections
  - *Escherichia coli*
  - *Klebsiella*
  - *Enterobacter*
  - *Citrobacter*
  - *Serratia*
  - *Edwardsiella*
  - *Proteus*
  - *Providencia*
  - *Morganella*

General characteristics

- Order Enterobacterales ≠ Family Enterobacteriaceae
- Enterobacterales refers to "enterics"
- They are normal flora of the gut, but not the other organs
- They can be found in human and animal intestine, environment (soil, water) and plant
- The order contains 7 family, 68 genus and >250 species
- All are gram-negative bacilli or coccobacilli
- They are facultative anaerobe and do not form spore
- They do not produce oxidase enzyme except *Plesiomonas* Oxidase +
- They all produce catalase except *Shigella dysenteriae* Catalase −
- They all can ferment glucose but may or may not produce gas
- Almost all of them can reduce nitrate to nitrite All most nitrate +
- They are motile *except Klebsiella*, *Shigella* and *Yersinia* 1 2 3



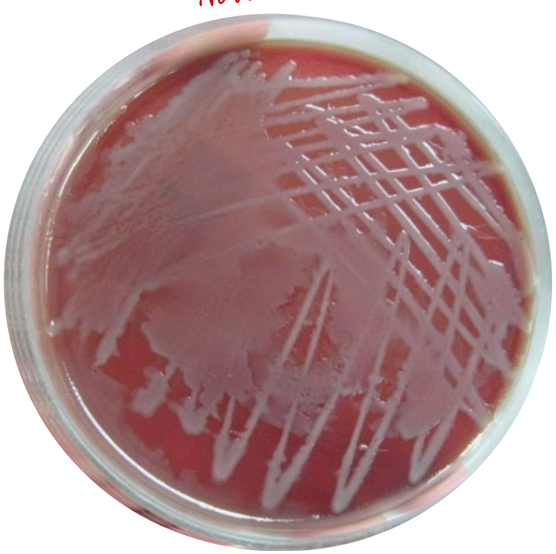
General characteristics (Cont.)

- Their colony morphology are middle-large and grey (กลม นูน ขอบเรียบ เป็นมัน ขนาด 2-3 mm. สีเทา γ-hemolysis)

- They has no remarkable colony morphology except *Proteus, Klebsiella, Enterobacter* and *Serratia.*

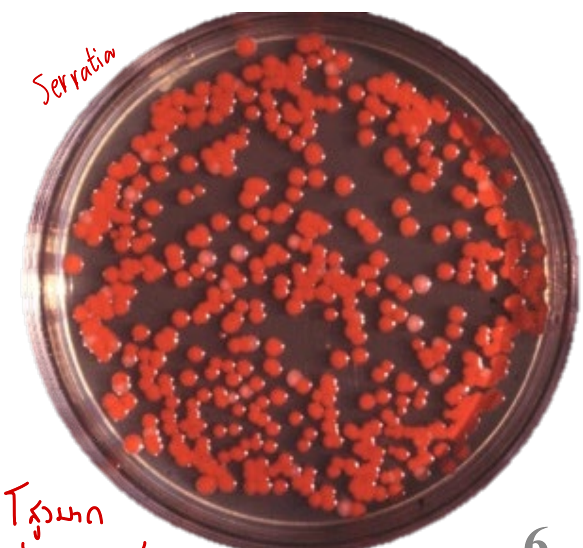
Normal colony

Protens



Klebsiella

Serratia



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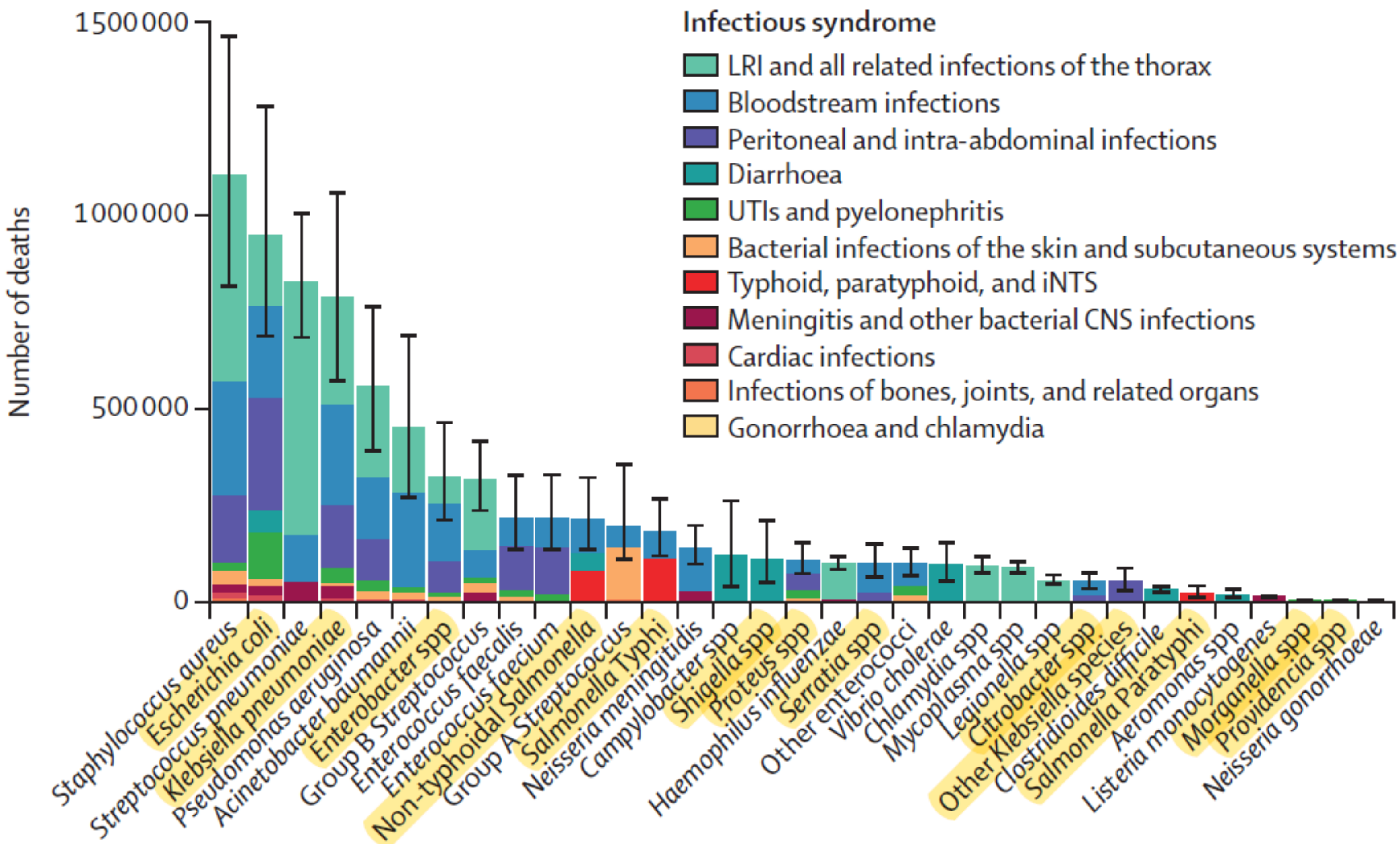
General characteristics (Cont.)

- They are the biggest group of medically important bacteria, most frequently isolated from patients and the most important causative agents of nosocomial infections.

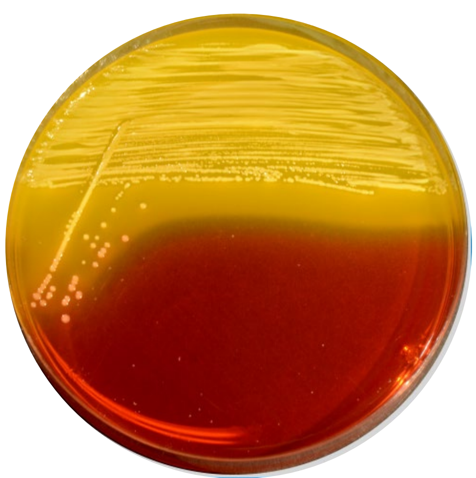


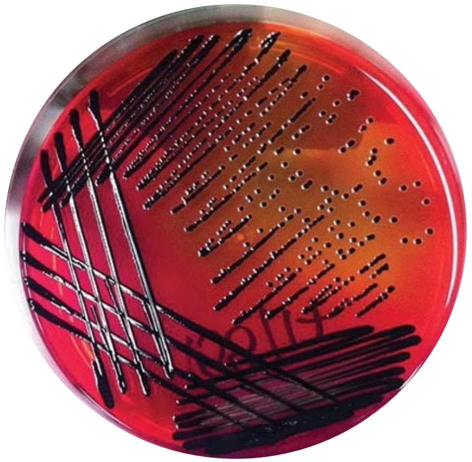
Global number of deaths by pathogen and infectious syndrome, 2019. Lancet. 2022; 400(10369): 2221-2248.

Cultivation of Enterobacterales

- Blood agar and MacConkey agar
  - 37 °C, ambient air
- Other selective media
  - XLD, SS, EMB
  - Contain specified sugar and pH indicator
  - H_2S can also be captured on XLD agar
  - 37 °C, ambient air

XLD agar





Enterobacterales Classification

Family	Genus	Species ในแต่ละ genus ที่พบได้บ่อยที่สุดในสิ่งส่งตรวจจากผู้ป่วย
1. Enterobacteriaceae	*Escherichia*	*E. coli*, *E. coli* inactive, Diarrheagenic *E. coli*
	Klebsiella	*K. pneumoniae*, *K. aerogenes**, *K. oxytoca*, *K. ozaenae*
	Enterobacter	*E. cloacae*
	Citrobacter	*C. freundii*, *C. koseri* (*diversus*)
	Cronobacter	*C. sakazakii**
	Salmonella	*S. enterica*
	Shigella	*S. dysenteriae*, *S. flexneri*, *S. boydii*, *S. sonnei*
2. Erwiniaceae	*Pantoea*	*P. agglomerans**
	Erwinia	ก่อโรคในพืช
3. Pectobacteriaceae	-	ก่อโรคในพืช
4. Yersiniaceae	*Serratia*	*S. marcescens*
	Yersinia#	*Y. pseudotuberculosis*, *Y. pestis*
5. Hafniaceae	*Edwardsiela*	*E. tarda*
	Hafnia#	*H. alvei*
6. Morganellaceae	*Proteus*	*P. mirabilis*, *P. vulgaris*, *P. penneri*
	Providencia	*P. alcalifaciens*, *P. rettgeri*, *P. stuartii*
	Morganella	*M. morganii*
7. Budviciaceae	*Leminorella*#	*L. grimontii*

PD Positive

* Genus เดิมคือ Enterobacter

(*Plesiomonas shigelloides* ไม่ถูกจัดอยู่ใน Family ใด ๆ)

พบได้ไม่บ่อยในประเทศไทย

Virulence and Antigenic Factors

Virulence factors

1. Ability to adhere, colonize, produce toxins, endotoxin, siderophore, invade tissues, resistant to serum killing

2. Harbor drug resistant genes on plasmid that give resistant to:

   2.1 Extended-spectrum cephalosporins (3rd and 4th generations)

   - Gene names: Ex. *bla*CTX-M, *bla*TEM, *bla*SHV
   - The produced enzymes is called Extended-spectrum β-lactamase (ESBLs)
   - Enterobacterales carrying ESBLs is called Extended-Spectrum Cephalosporin-Resistant Enterobacterales (ESCR-E)

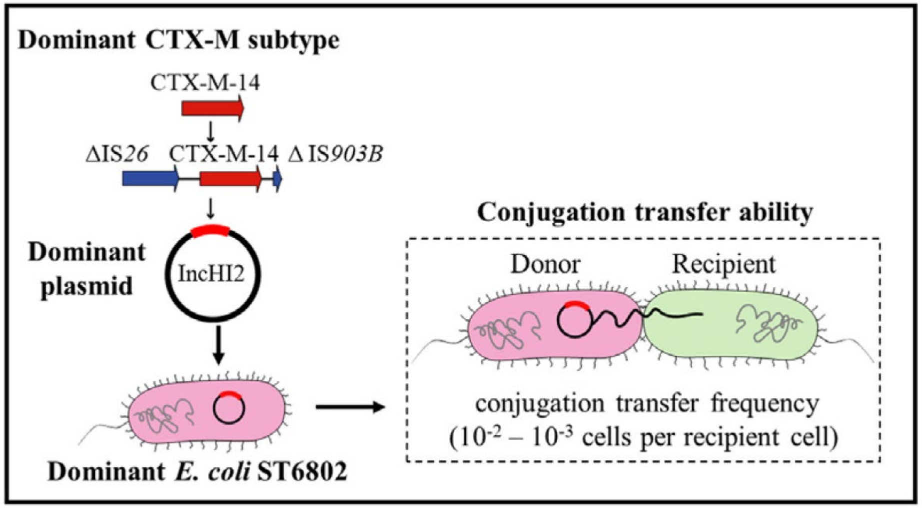
Virulence and Antigenic Factors

Virulence factors

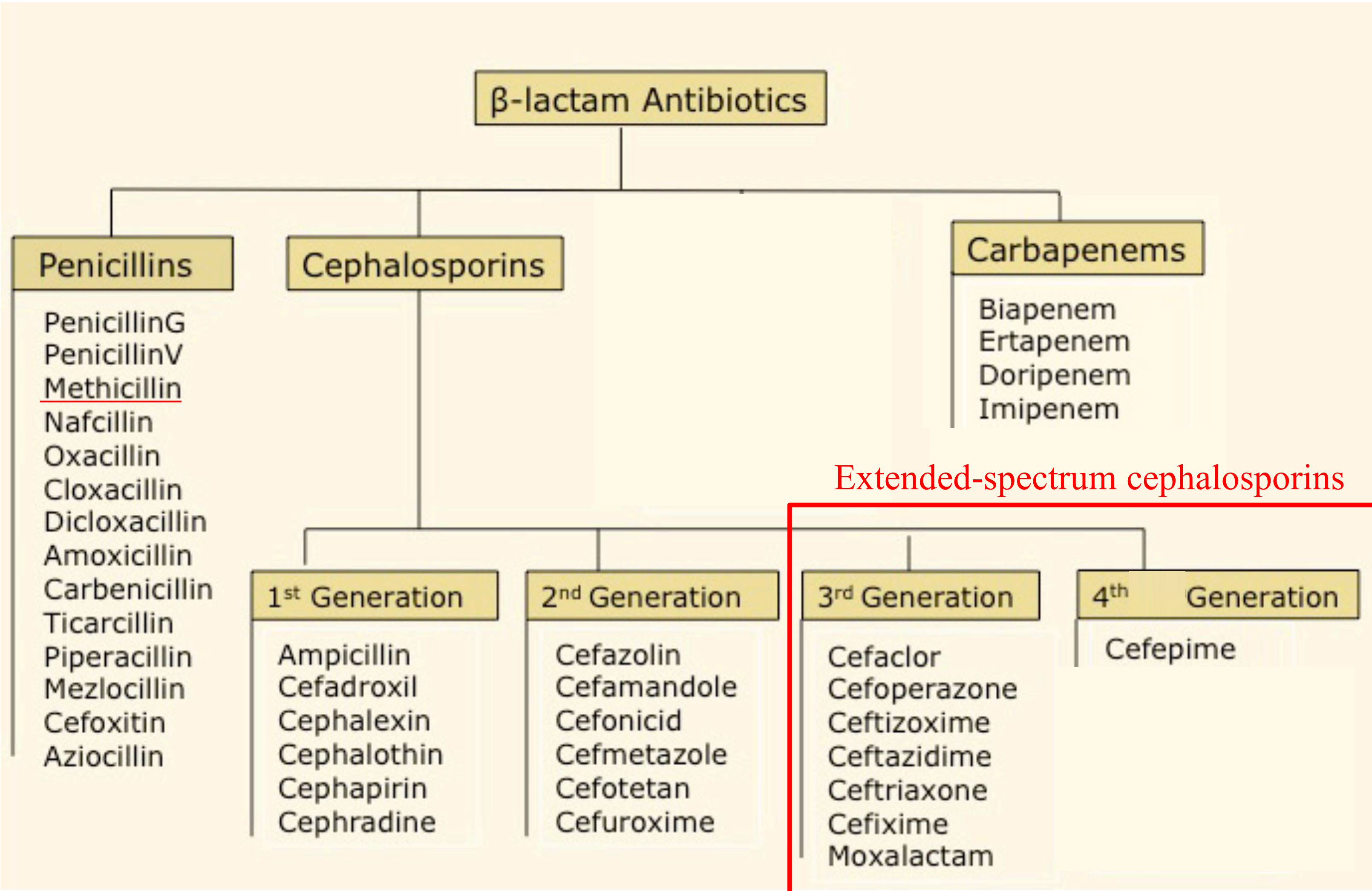
2. Harbor drug resistant genes on plasmid that give resistant to:

2.2 Carbapenems

- Gene names: Ex. *bla*NDM, *bla*OXA, *bla*KPC, *bla*IMP, and *bla*VIM
- The produced enzymes is called Carbapenemase
- Enterobacterales with resistant to carbapenems is called Carbapenem-Resistant Enterobacterales (CRE)
- Gram negative bacteria that produce carbapenemase is called carbapenemase producing organism (CRO)



Virulence and Antigenic Factors



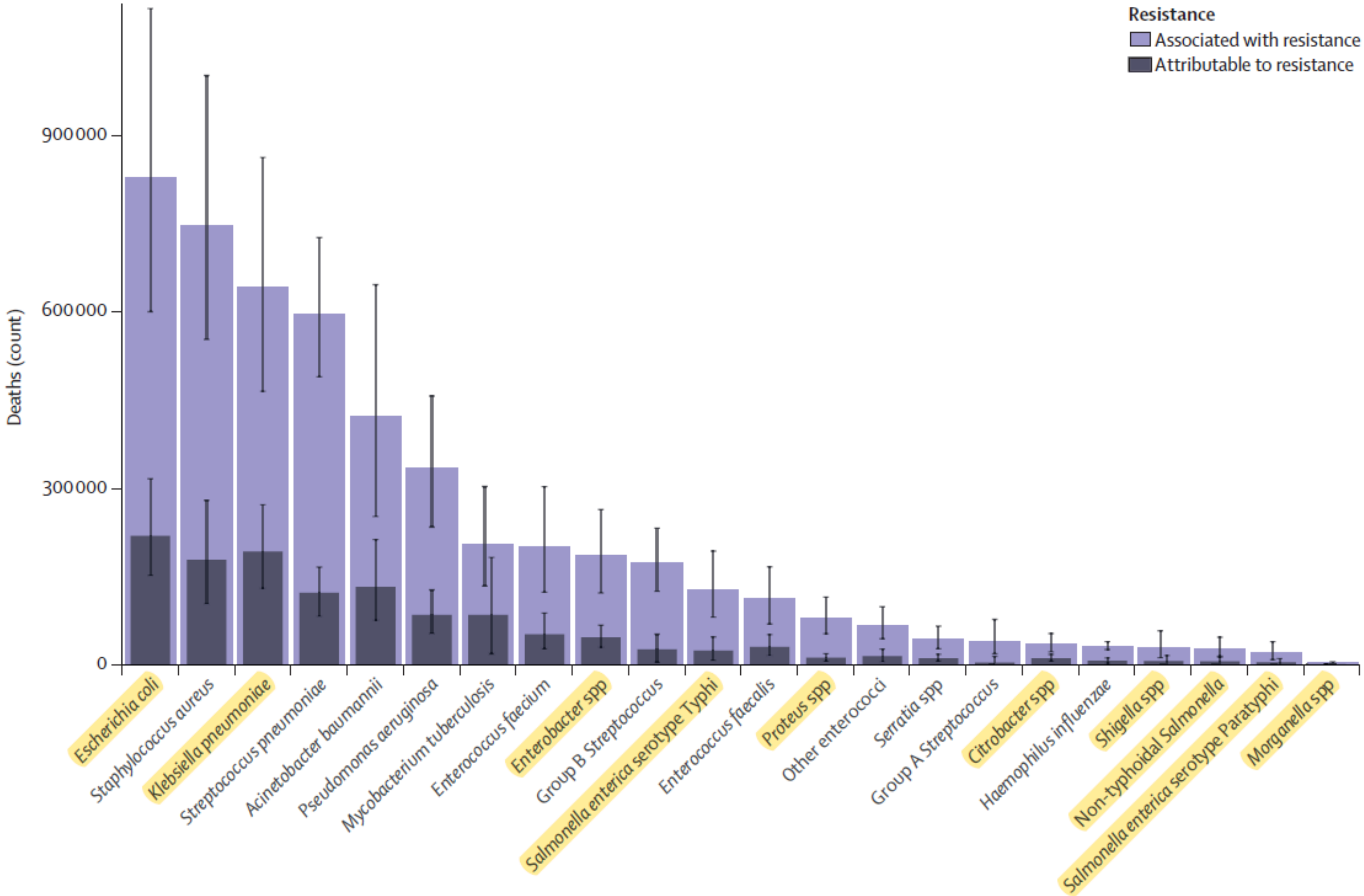
Virulence and Antigenic Factors

WHO Bacterial Priority Pathogens List, 2024

Bacterial pathogens of public health importance to guide research, development and strategies to prevent and control antimicrobial resistance

Fig 4. WHO Bacterial Priority Pathogens List, 2024

Virulence and Antigenic Factors

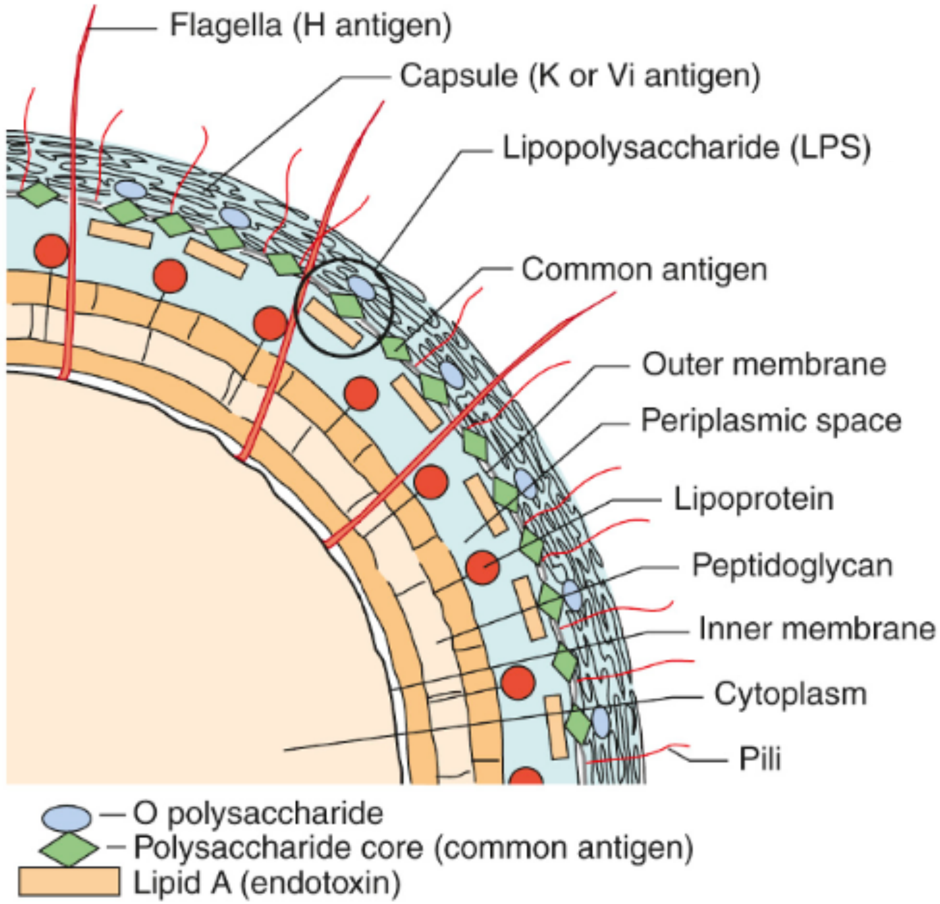


Global deaths (counts) attributable to and associated with bacterial antimicrobial resistance by pathogen, 2019

Lancet. 2022; 399(10325): 629-655

Virulence and Antigenic Factors

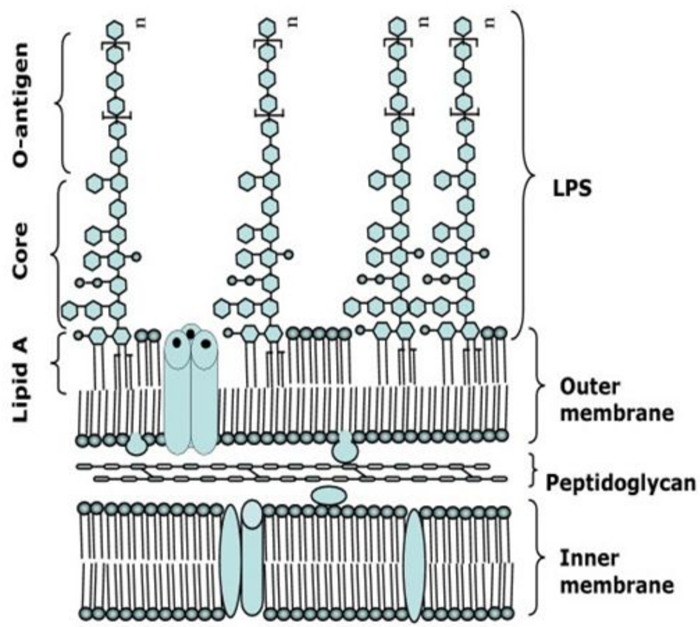
Cell wall of Enterobacterales



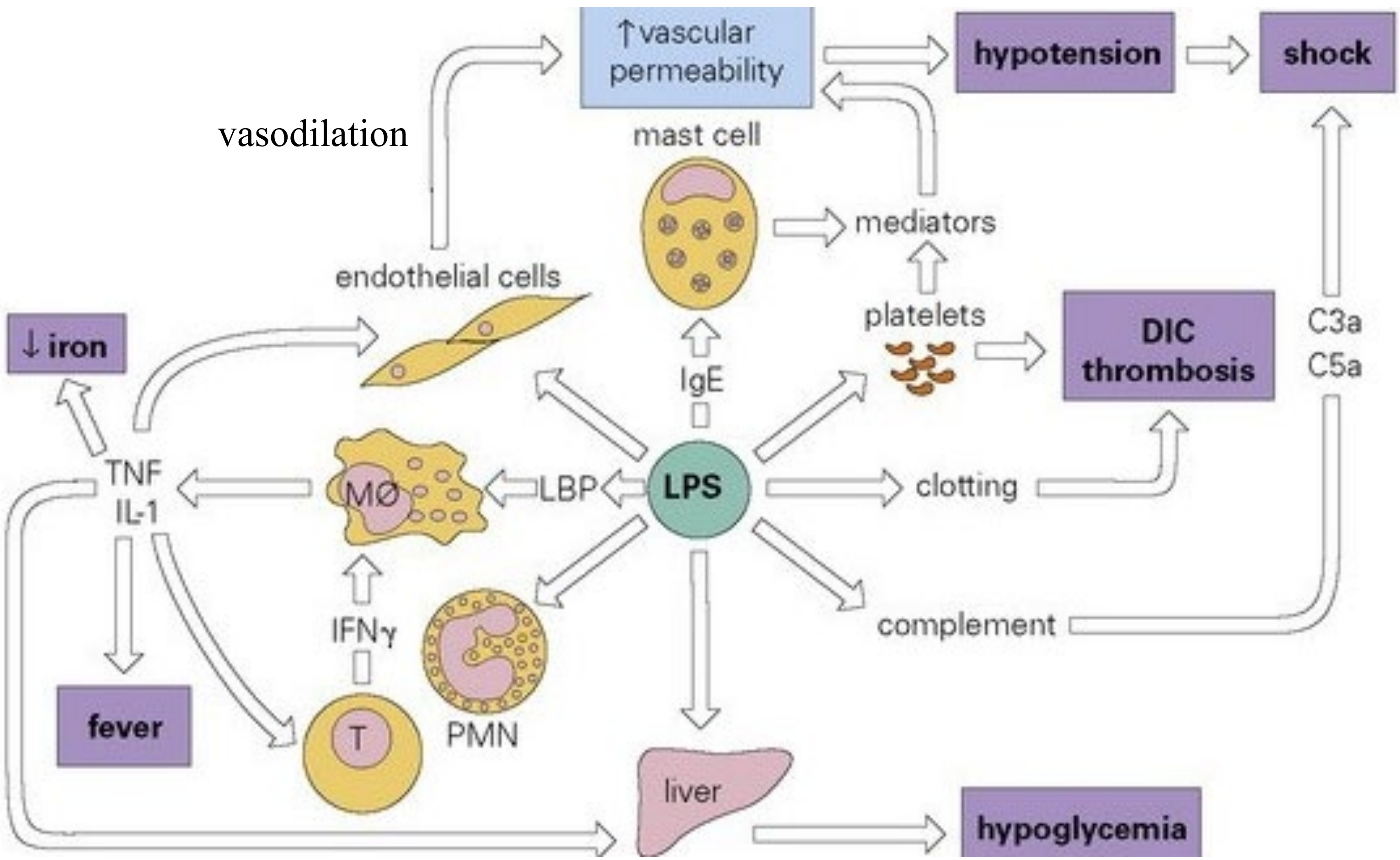
Lipopolysaccharide

(LPS) consists of

1. - O polysaccharide
2. - Polysaccharide core
3. - Lipid A (endotoxin)



Virulence and Antigenic Factors



Virulence and Antigenic Factors

Antigenic factors

1. O (somatic) antigen   species เดียวกัน O ต่างกันได้

   : a heat-stable antigen on cell wall

   : found in all Enterobacterales

2. H (flagella) antigen

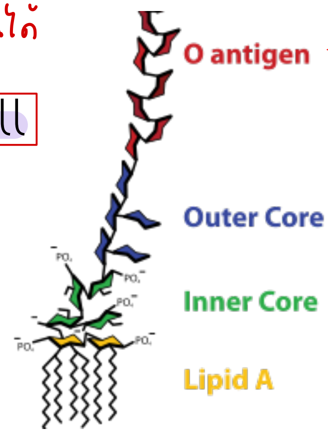
   : a heat-labile antigen found on surface of flagella, mostly flagellin

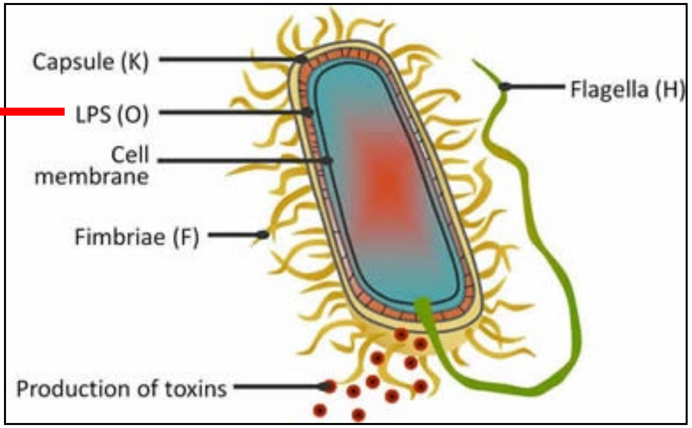
   : found in some Enterobacterales

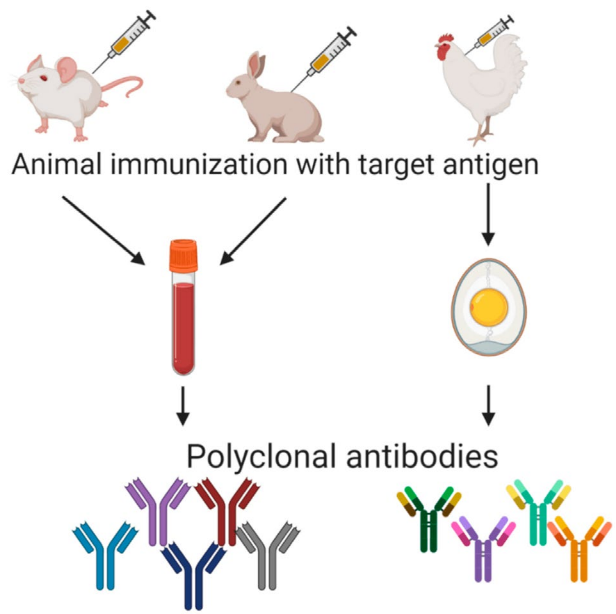
3. K (capsular) antigen

   : a heat-labile polysaccharide

   : found in some Enterobacterales e.g. *E. coli, S.* Enterica serotype Typhi





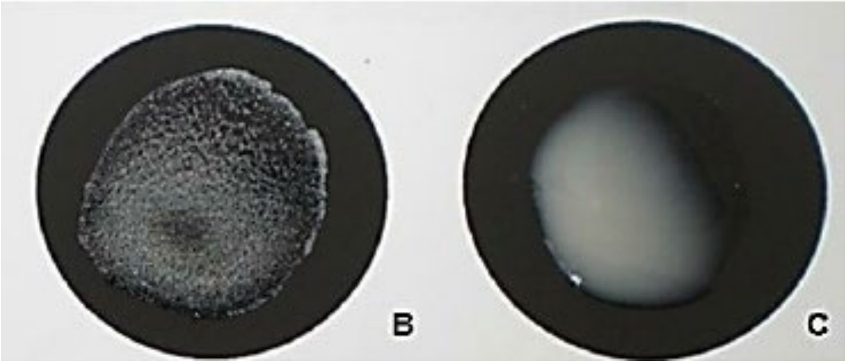


Virulence and Antigenic Factors

Antigenic factors (cont.)

- O, H, K-Ag can be used for identification of different serologic groups (serogroup)

- K-Ag may mask O-Ag, boiling at 100 °C for 10 min is helpful



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Clinical significance

Enterobacterales can be responsible for a large number of opportunistic infections when introduce into inappropriate body sites. Based on clinical infections, they can be divided into

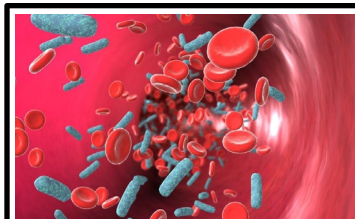
1. Opportunistic pathogens e.g., *E. coli*, *Klebsiella* spp., *Enterobacter* spp., *Proteus* spp. & etc. which can cause UTI, sepsis, pneumonia, meningitis, wound infections and abscesses, which mostly is hospital-acquired (nosocomial) infection

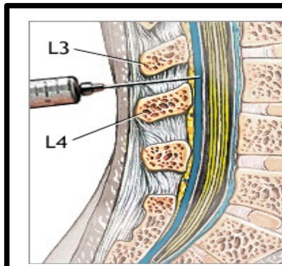
ก่อโรคในคนที่มีสุขภาพแข็งแรง

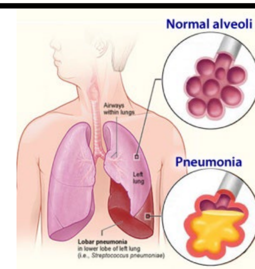
2. True pathogens i.e.
   1. Diarrheagenic *E. coli*
   2. *Salmonella* spp.
   3. *Shigella* spp.
   4. *Yersinia pestis*

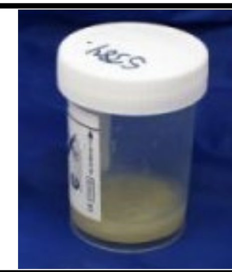
Bacterial species and Diseases

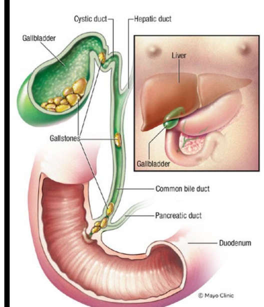
Bacterial Species	Diseases
Escherichia coli	Bacteriuria, septicemia, neonatal sepsis, meningitis, diarrheal syndrome
Shigella spp.	Diarrhea, dysentery
Edwardsiella spp.	Diarrhea, wound infection, septicemia, meningitis, enteric fever
Salmonella spp.	Septicemia, enteric fever, diarrhea
Citrobacter spp.	Opportunistic and hospital-acquired infections (wound, urinary)
Klebsiella spp.	Bacteriuria, pneumonia, septicemia
Enterobacter spp.	Opportunistic and hospital-acquired infection, wound infections, septicemia, bacteriuria
Serratia spp.	Opportunistic and hospital-acquired infection, wound infections, septicemia, bacteriuria
Proteus spp.	Bacteriuria, wound infection, septicemia
Providencia spp.	Opportunistic and hospital-acquired infection, wound infections, septicemia, bacteriuria
Morganella spp.	Opportunistic and hospital-acquired infections



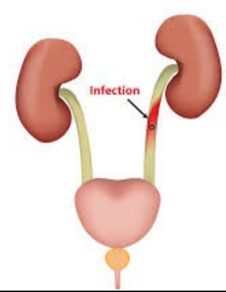








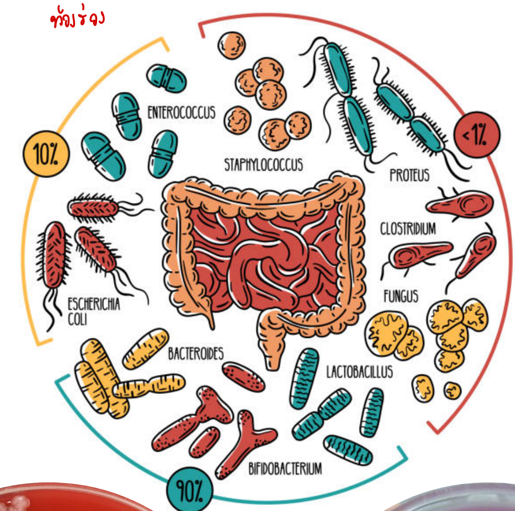


Escherichia coli

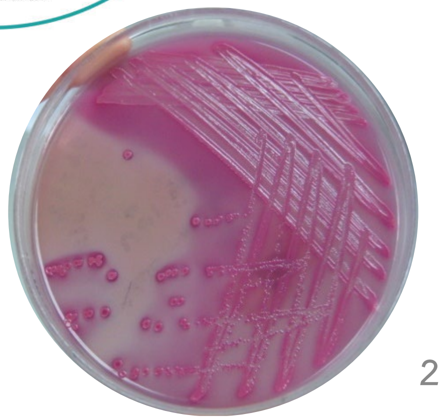
E. coli is a Gram-negative, rod-shaped, non-spore forming bacterium belonging to the Enterobacteriaceae family. It is widely distributed in the intestines of humans and animals. It can cause UTI, sepsis, pneumonia, CNS infection and diarrhea.

ทางเดินปัสสาวะ ติดเชื้อในกระแสเลือด ปอด สมอง ท้องร่วง

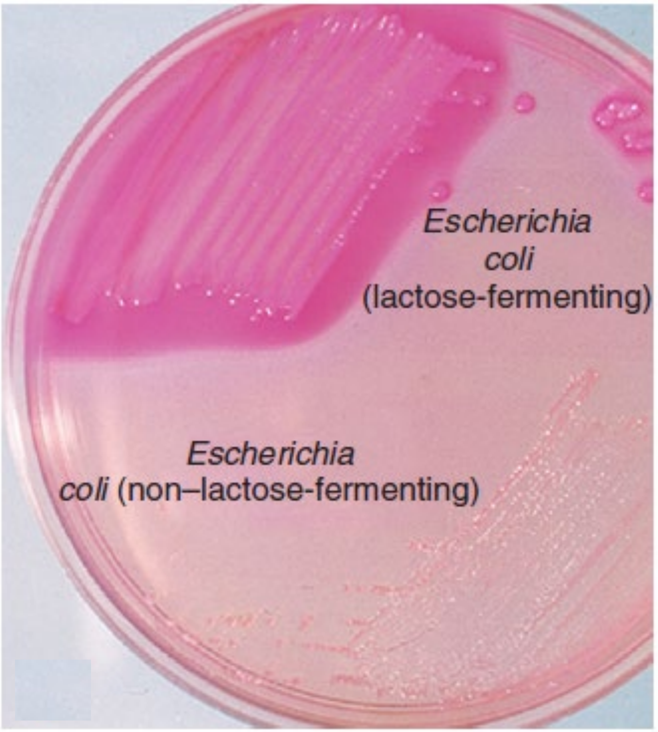
Most strains are motile and produce adhesive fimbriae, sex pili, O, H and K antigens.

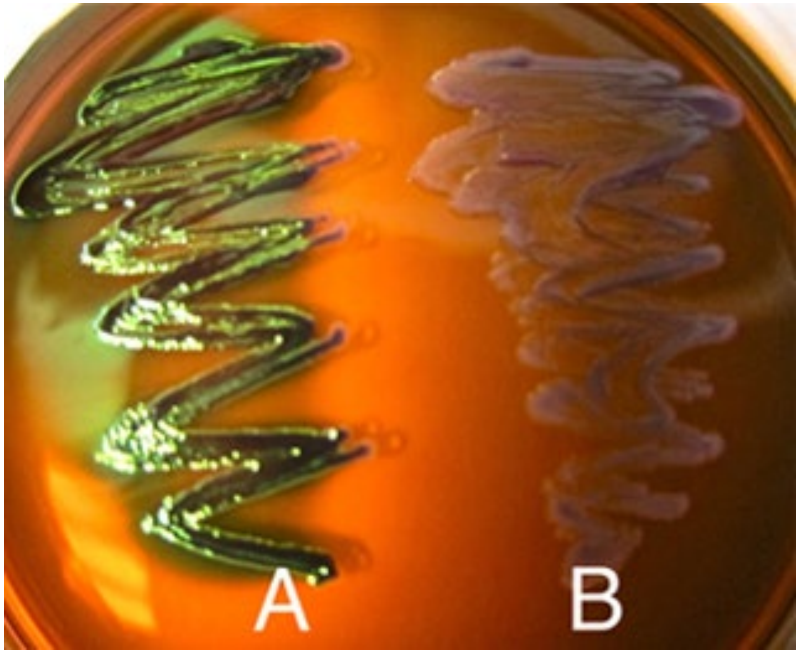


Serotyping of O and H antigens are useful for identification of strains associated with serious enteric diseases.



Escherichia coli (Cont.)





MacConkey agar :

Lactose fermenter

Eosin Methylene Blue (EMB)

agar : Green Metallic Sheen

Escherichia coli (Cont.)

Diseases

1. Diarrheagenic (True pathogen)
   - Enterotoxigenic *E. coli* (ETEC)
   - Shiga Toxin Producing *E. coli* (STEC)
     - Enterohemorrhagic *E. coli* (EHEC)
   - Enteropathogenic *E. coli* (EPEC)
   - Enteroinvasive *E. coli* (EIEC)
   - Enteroaggregative *E. coli* (EAEC)

2. Extraintestinal infections (Opportunistic infection): UTI, sepsis, meningitis



1. Enterotoxigenic *Escherichia coli* (ETEC)

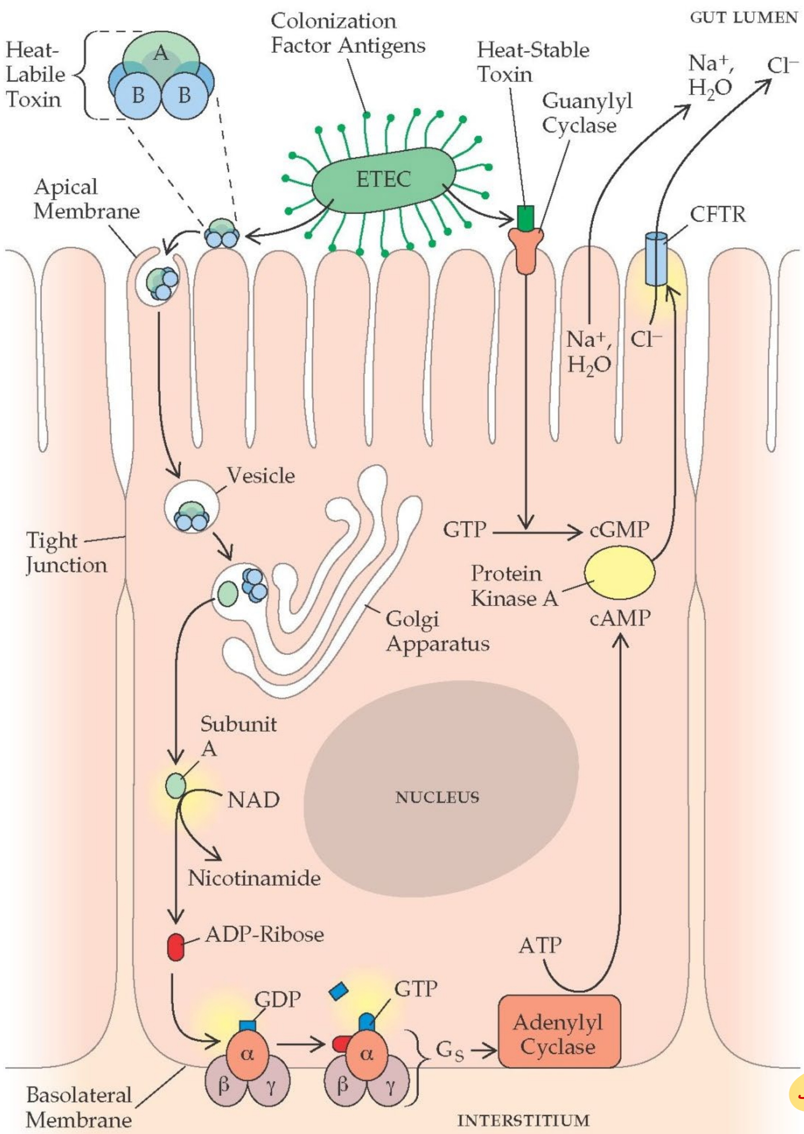
General Information

- ETEC causes watery diarrhea in children in tropical and subtropical area, especially in developing countries.

- It is also the major cause of traveler's diarrhea, the symptoms when the patient has 3 or more bouts of loose, watery poo in 24 hours. Most cases are mild, but for some people it is severe. It usually happen in the first week of travel. The symptoms last for 3 to 5 days and usually get better without treatment.

หายเองได้

1. Enterotoxigenic *Escherichia coli* (ETEC)



Virulence factors: colonizing factors (CFs), heat labile toxin (LT), heat stable toxin (ST)

Pathogenicity:

- ETEC bind to small bowel epithelium using surface protein call "colonization factor" (CF)
- It produce two enterotoxins (plasmid encoded)
  - heat-labile toxin (LT)
  - heat-stable toxin (ST)
- LT is similar to cholera toxin from *V. chlolerae*
- Both toxins activate formation of cyclic AMP, which cause secretion of electrolytes (Cl^-) and fluid into intestinal lumen.

* toxins กระตุ้นการเกิด cAMP ทำให้ หลั่ง Cl^- และของเหลวเข้าสู่ intestinal lumen

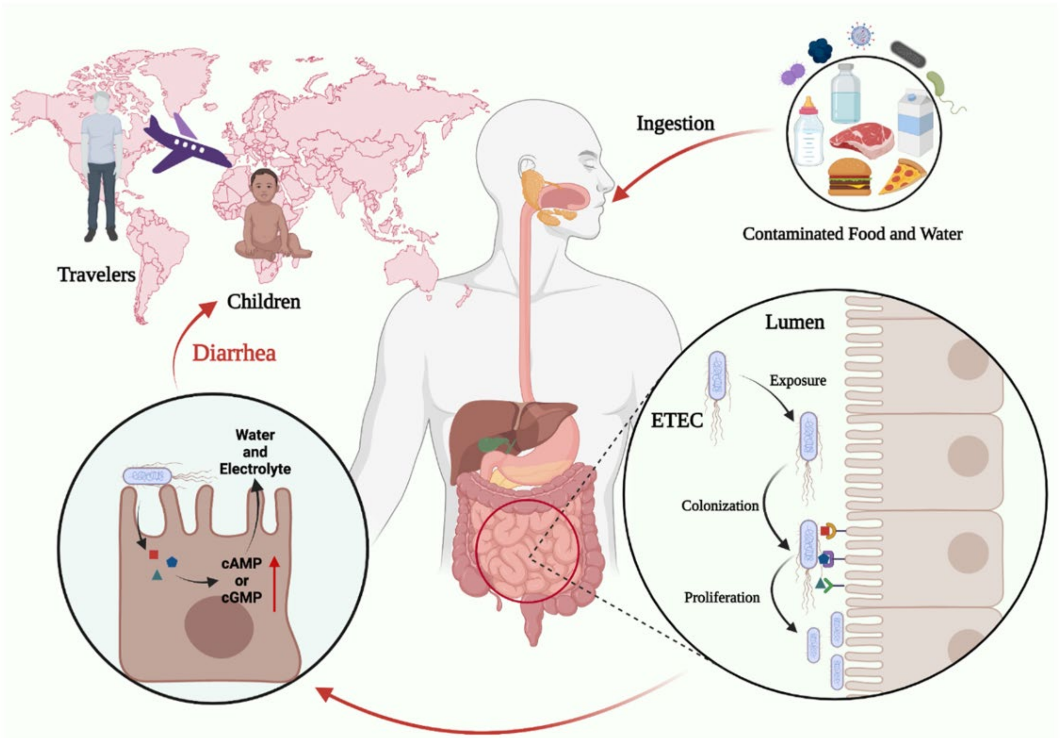
1. Enterotoxigenic *Escherichia coli* (ETEC)

Infective dose: 10^6 – 10^{10} CFUs

Symptoms: watery & non-bloody diarrhea, but abdominal cramps, nausea, vomiting and fever are less common. No mucosal invasion and the symptom is self-limited

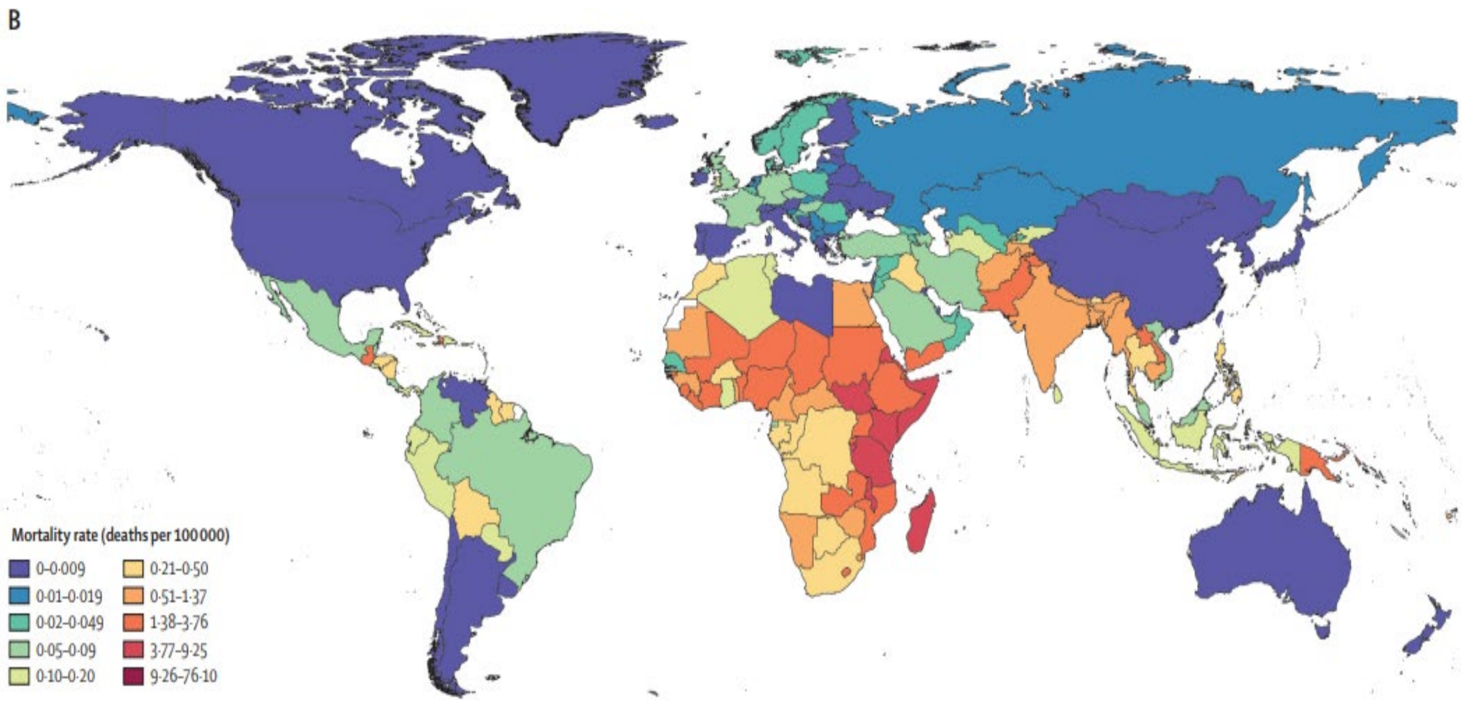
แล้งๆ

Transmission: ETEC is mainly spread via food and water, but person-to-person transmission was also reported. Asymptomatic carriers is common, but NO animal reservoirs are reported.



1. Enterotoxigenic *Escherichia coli* (ETEC)

Epidemiology: ETEC infections are prevalent in developing countries, especially in areas with inadequate water sanitation such as Africa, South Asia and South-east Asia including Thailand



ETEC diarrhea mortality rate per 100,000 people in 2016 for all ages

https://www.thelancet.com/journals/laninf/article/PIIS1473-3099(18)30475-4/fulltext

1. Enterotoxigenic *Escherichia coli* (ETEC)

Travelers' Health

Search

Travelers Health > CDC Yellow Book > Table of Contents > Escherichia coli, Diarrheagenic

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CDC YELLOW BOOK

Escherichia coli, Diarrheagenic

CDC Yellow Book 2024

Travel-Associated Infections & Diseases

Author(s): Jennifer Collins, Danielle Tack,

On This Page

- Infectious Agent
- Transmission
- Epidemiology
- Clinical Presentation

Epidemiology

The 2010 World Health Organization (WHO) Global Burden of Foodborne Diseases report estimated ≈111 million illnesses and ≈63,000 deaths caused by diarrheagenic *E. coli* globally each year. Rates of infection vary by region, and certain types of diarrheagenic *E. coli* infections, mainly ETEC, are associated with travel to low- and middle-income countries. The incidence of travel-associated diarrhea caused by *E. coli* is likely underestimated because many travelers do not seek medical care or have stool testing performed, particularly if diarrhea is non-bloody, as commonly occurs with ETEC infection. Moreover, many clinical laboratories do not use methods that can detect diarrheagenic *E. coli* other than STEC in stool samples.

Risk for travelers' diarrhea can be divided into 3 levels, according to the destination country. Low-risk countries include Australia, Canada, Greenland, Japan, New Zealand, the United States, and countries in northern and western Europe. Intermediate-risk countries include Argentina, Brazil, Chile, Morocco, Portugal, South Africa, Thailand (in Bangkok, Chiang Mai, and Phuket; risk to travelers going to rural areas is likely greater), Uruguay, and most countries in the Caribbean, eastern Europe, and the Middle East. High-risk countries include Afghanistan, Burma (Myanmar), the Indian subcontinent, Indonesia, Iran, Malaysia, Mexico, Papua New Guinea, most countries in Africa, and countries in Central America and northern South America, including Bolivia and Paraguay.

STEC infections are most commonly reported in industrialized countries, and ≈85% of STEC infections among international travelers are caused by non-O157 serotypes. Additional information about travelers' diarrhea is available in Sec. 2, Ch. 6, Travelers' Diarrhea.

Last access: 10 February 2025

Diagnosis: PCR (*elt* and *estA* genes)

Treatment: Rehydration and antibiotics such as azithromycin and ciprofloxacin

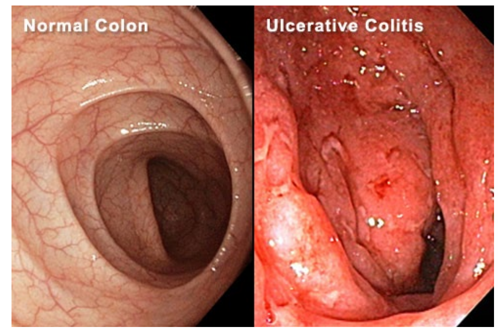
2. Shiga toxin producing *Escherichia coli* (STEC)

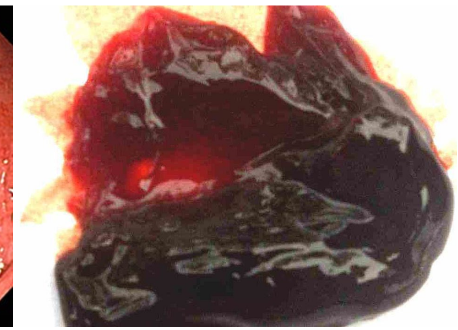
General Characteristic

ทำให้เกิด

ลำไส้ใหญ่อักเสบเรื้อรัง

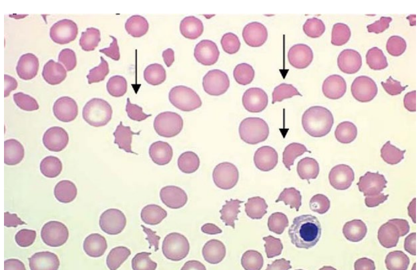
- STEC causes mild watery to bloody diarrhea, colitis, abdominal cramps, mild fever
- All STEC strains produce Shiga toxin. If the strain has additional virulence factors resulting in $\boxed{\text{hemorrhagic colitis}}$ (bloody diarrhea) and, in some cases, life-threatening $\boxed{\text{hemolytic uremic syndrome}}$ (HUS), that strain is Enterohemorrhagic *E. coli* (EHEC)





- The most common strain is *E. coli* O157:H7, which $\boxed{\text{produce both Stx1 and Stx2.}}$
- 5-10% of children younger than 10 years develops HUS
- HUS = acute renal failure + low platelet + microangiopathic hemolytic anemia

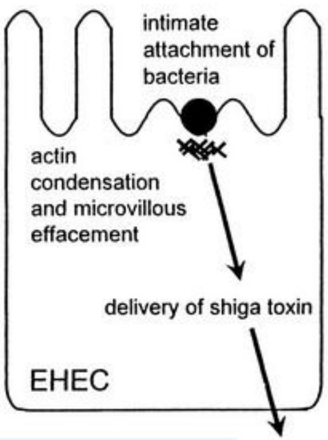
1 ไตวายเฉียบพลัน 2 เกล็ดเลือดต่ำ 3 เม็ดเลือดแดงแตก



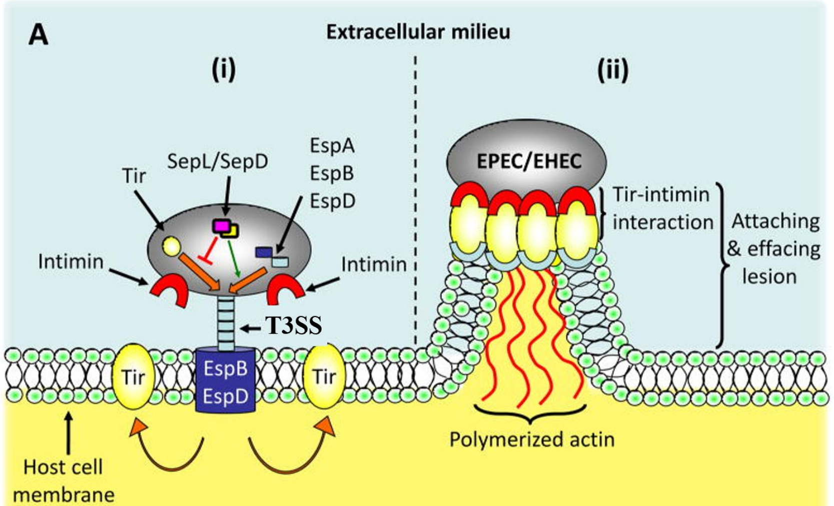
2. Shiga toxin producing *Escherichia coli* (STEC)

Virulence factors:

1. Shiga Toxins (Stx): Stx1 and Stx2, inhibit protein synthesis leading to cell death
2. Intimin: Encoded by the *eae* gene, mediates adherence to intestinal epithelial cells. It enables STEC to form "attaching and effacing" lesions, damaging the intestinal lining.



3. Type III Secretion System (T3SS): This system enables STEC to inject effector proteins into host cells, which subvert cellular processes and promote bacterial colonization. The T3SS is encoded on the locus of enterocyte effacement (LEE) pathogenicity island.

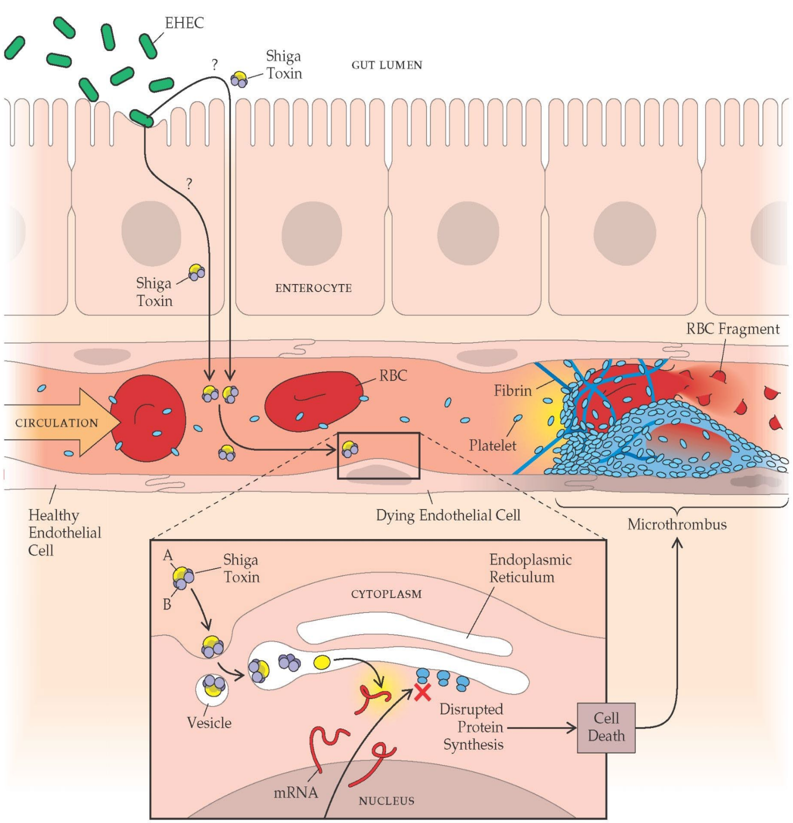


4. Hemolysins: STEC may produce enterohemolysins, such as EhxA

2. Shiga toxin producing *Escherichia coli* (STEC)

Pathogenicity

- Shiga toxin (Stx1) / Stx2 bind to receptor (Gb3) on enterocyte 1 and renal endothelial cells 2.
  Then get into the cells and inhibit 28s ribosome causing inhibition of protein synthesis leading to cell death.



- Enterocyte cell death leads to mucosal lesion and mucous bloody diarrhea
- Vessel endothelial cell death causes activation of platelets and thrombin deposition, which block circulation.

  สร้าง สร้าง
- Squeezing of the blood through to the blocked vessels lead to hemolytic anemia
- Destruction of glomerular endothelial cells kidney tissue causes decrease GFR and renal failure.

2. Shiga toxin producing *Escherichia coli* (STEC)

Infective does: < 100 CFUs

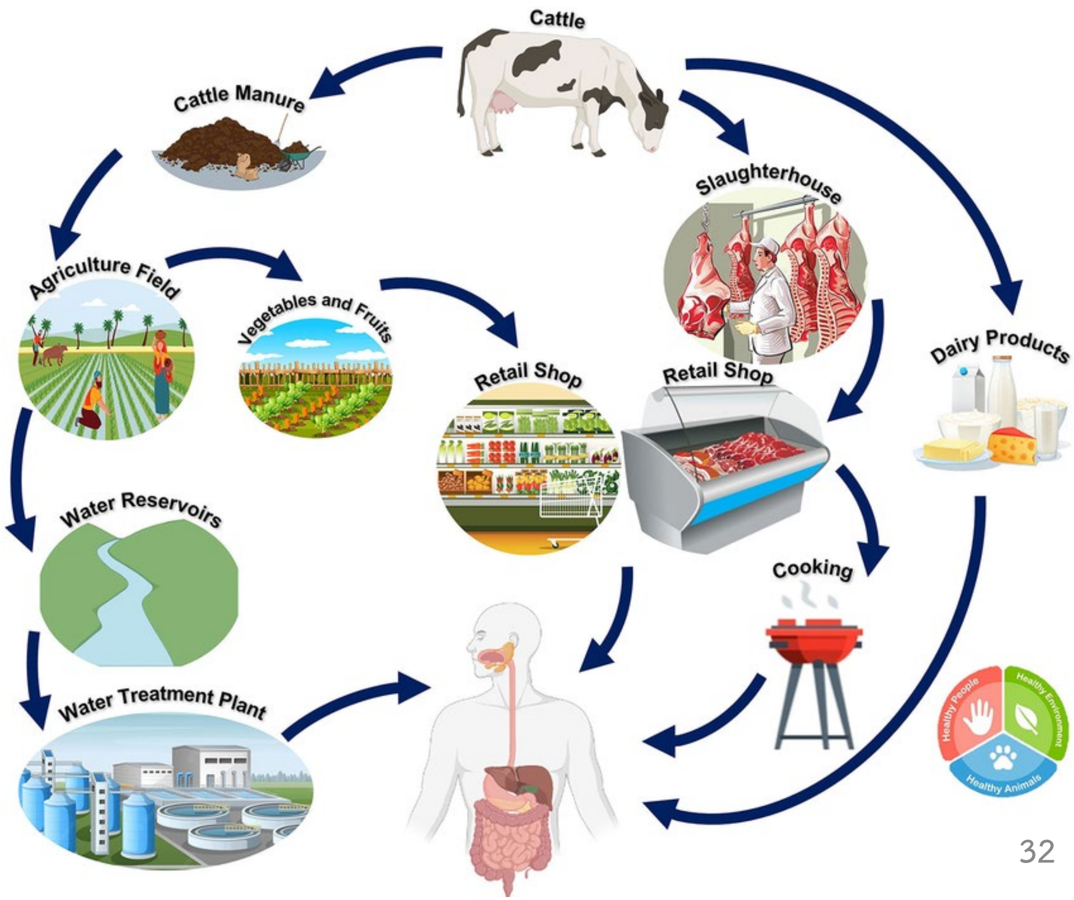
Symptom: Typically begins with mild fever, mild watery diarrhea, abdominal cramps and, within a few days, may progress to bloody diarrhea. In severe cases it can lead to HUS.!

Transmission:

- Infection is caused by consumption of contaminated undercooked meat, water, milk, fruit juice and fresh vegetables.

- Person-to-person spread was reported.

- Cattle is believed to be reservoir host of STEC/EHEC



2. Shiga toxin producing *Escherichia coli* (STEC)

Epidemiology:
- STEC are mostly reported in North America, Europe, Japan, and Australia, where foodborne illness surveillance is extensive. The WHO estimates about 2.8 million cases of STEC infections annually.

- While data from developing regions are more limited, STEC are also present in Africa, Asia, and Latin America, though likely underreported due to limited resources for detection and reporting.

Proportional circles map of major outbreaks of EHEC reported during 1985–2017. (https://www.mdpi.com/2072-6651/13/5/306)

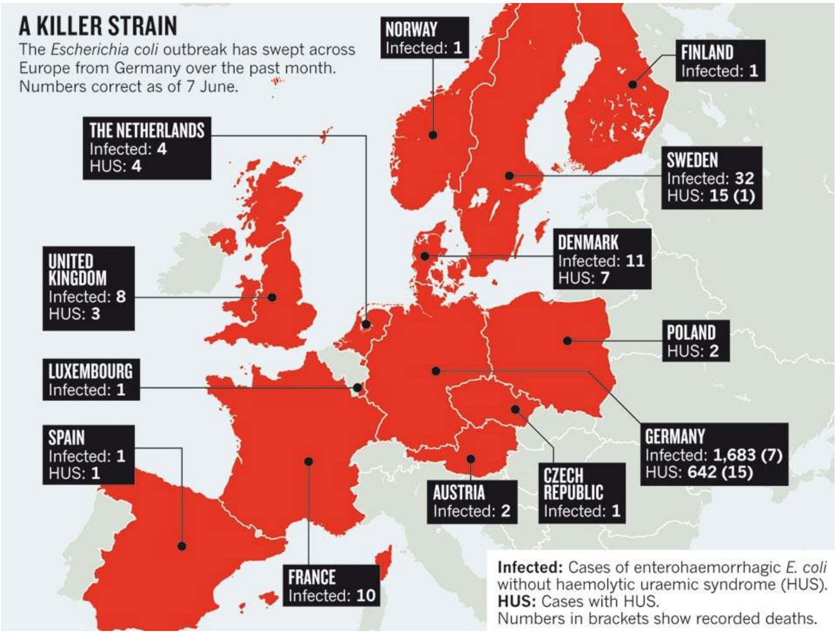
- Diagnosis:
  - Screening: sorbitol MacConkey agar
  - Confirm : PCR (stx1 & stx2 gene)

- Treatment: Rehydration and supportive care are primary treatments, as antibiotics are generally, not recommended (they may increase toxin release and risk of HUS).

2. Shiga toxin producing *Escherichia coli* (STEC)

Outbreak of Diarrheagenic *E. coli* in Germany, 2011

- *E. coli* O104:H4
- This strain has characteristics of EHEC and EAEC
- 3,842 persons were infected
- 855 cases developed HUS
- 53 deaths
- the outbreak was caused by



2. Shiga toxin producing *Escherichia coli* (STEC)

CDC

E. coli Infection (*Escherichia coli*)

EXPLORE TOPICS

SEARCH

OCTOBER 30, 2024

Investigation Update: *E. coli* Outbreak, Onions Served at McDonald's - October 30, 2024

KEY POINTS

Epidemiologic and traceback data show that fresh, slivered onions served at McDonald's are the likely source of this outbreak. Onions were pulled from affected McDonald's locations and Taylor Farms recalled yellow slivered onions sent to other food service operators.

Notice | Locations | Timeline

Demographics	Information
Age (n=90)	Range from 13 to 88 years Median age of 30
Sex (n=90)	41% female 59% male
Race (n=82)	96% White 3% African American/Black 0% Native American or Alaska Native 0% Asian 0% Native Hawaiian or other Pacific Islander 1% reported more than one race
Ethnicity (n=79)	92% non-Hispanic 8% Hispanic

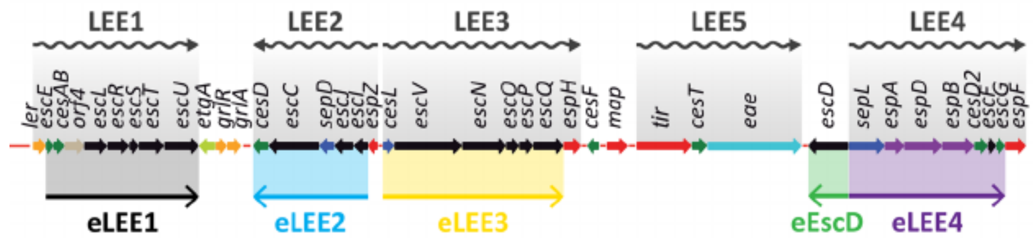
How disease detectives' quick work traced deadly E. coli outbreak to McDonald's Quarter Pounders

By Brenda Goodman
Sunday, November 3, 2024 4:53AM

EHEC outbreak in USA
83 infected
27 hospitalized
2 developed HUS
1 death

3. Enteropathogenic *Escherichia coli* (EPEC)

Enteropahogenic *E. coli* (EPEC)



- Cause infantile and non-bloody diarrhea, fever and vomiting.

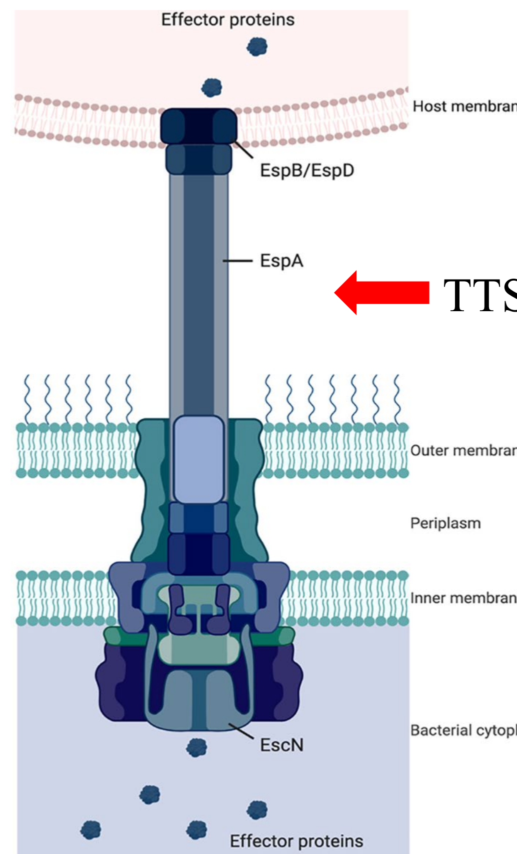
- Has a cluster of virulence genes on chromosome (pathogenic island) called locus of enterocyte effacement (LEE), which encodes Type III secretion (TTS) system.

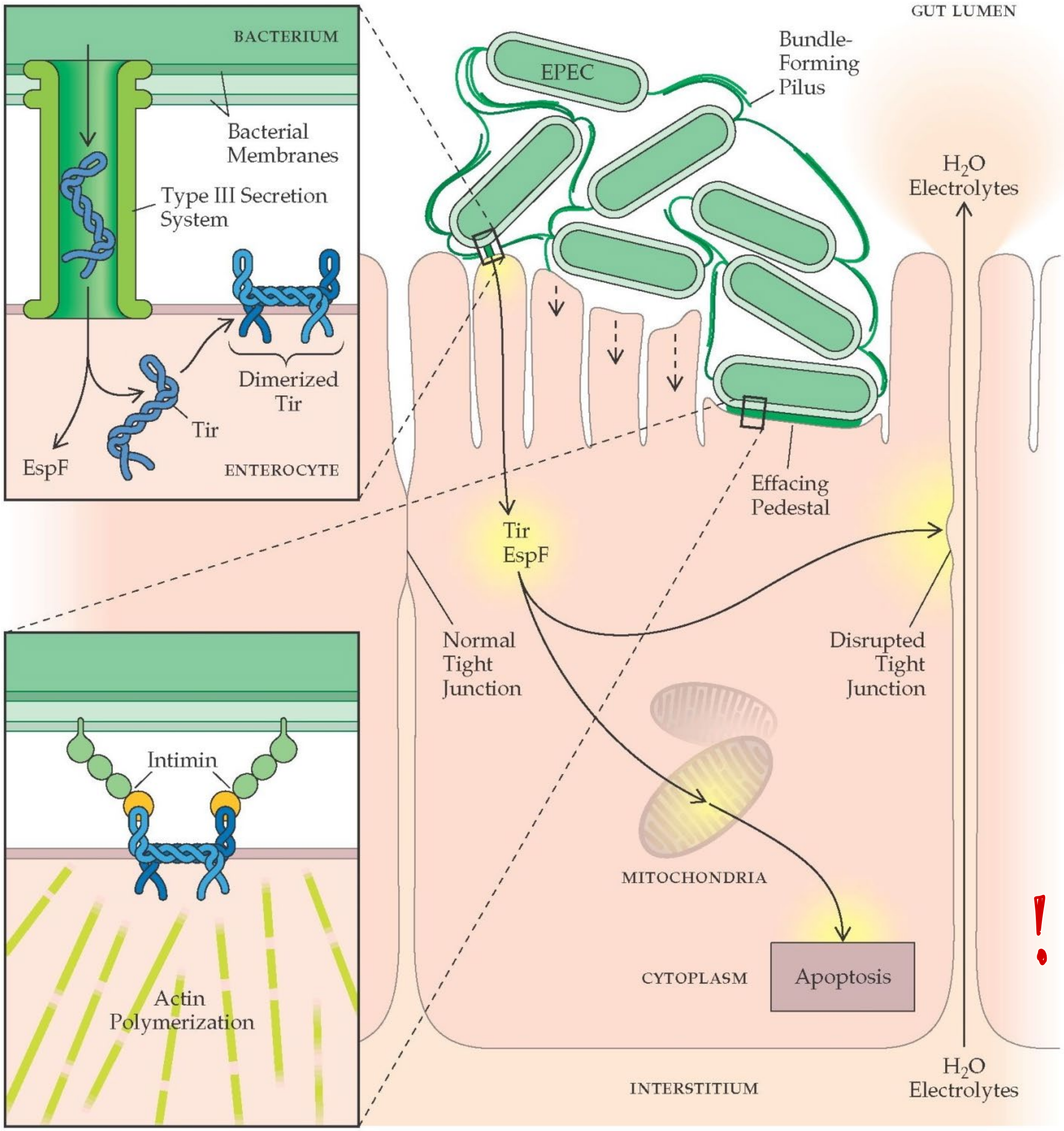
- Infection is initiated by bacterial attachment to small ! intestinal epithelial cells, then effacement of microvilli.

bac จับ แล้ว ปล่อย Protein เข้า epithelial cell ของ Host โดย T3SS

- After attachment, the bacteria secrete protein into host epithelial cells by Type III secretion system.

* - One of the secreted protein is translocated intimin receptor (Tir)





- Tir is inserted into host cell membrane and functions as receptor for bacterial membrane protein, intimin

- Binding of intimin to Tir results in polymerization of actin, accumulation of cytoskeleton beneath the attached bacteria

- Loss of cell surface integrity and cell death leads to watery diarrhea

! - * EPEC do not produce exotoxin * !

- Diagnosis: PCR (gene *espB* on LEE pathogenicity island)

4. Enteroinvasive *Escherichia coli* (EIEC)

Enteroinvasive *E. coli* (EIEC)

อาการคล้าย shigella Biochem คล้าย shigella

- EIEC is less common than ETEC and EPEC
- The illness is very similar to *Shigella* spp. infection !
- The symptoms are watery (and may be bloody) ! diarrhea, fever, abdominal cramps and malaise
- Colony morphology and biochemical testing are very similar to *Shigella* spp. !
- Infective dose is 10^6 cells (for *Shigella* is 10^2 cells)
- EIEC produce dysentery (โรคบิด) with direct penetration and destruction of large intestinal mucosa.
- The bacteria invade into epithelium then lyse the phagocytic vacuole and replicate in cytoplasm
- Epithelial destruction can progress to ulceration. !

5. Enteroaggregative *Escherichia coli* (EAEC)

Enteroaggregative *E. coli* (EAEC)

- A heterogeneous collection of strains characterized by their autoagglutination in a "stacked-brick" arrangement over epithelium cells of small intestine.
- After binding, cytokine release is stimulated which results in neutrophil recruitment
- Symptoms: watery diarrhea with inflammatory cells, fever, nausea, vomiting and abdominal pain.
- The disease can be acute or progress to persistent diarrhea, particularly in children and HIV-infected person !
- The prevalence of the disease is unclear because the molecular marker has not been discovered.
- Diagnosis: PCR (The plasmid-encoded transcriptional activator *aggR*, pathogenicity island associated gene *aaiC*, and the plasmid encoded dispersin transporter *aatA*)

Diarrheagenic *E. coli*

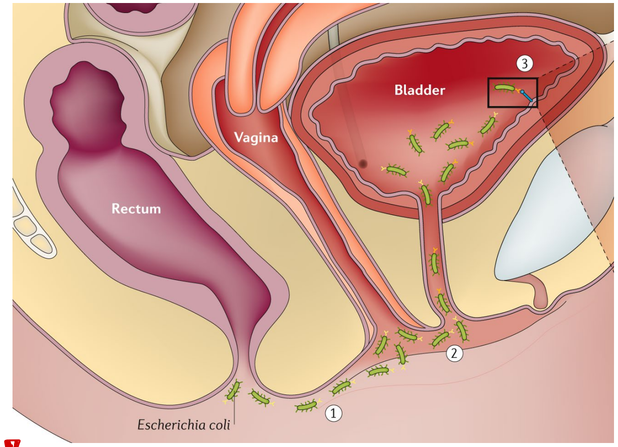
Properties / symptoms	ETEC	EHEC	EPEC	EIEC	EAEC
Intestine involved	Small	Colon	Small	Lower small, colon	Colon
Toxin	LT/ST	Shiga 1&2 (Vero)	-	-	ESAT1
Intimin (*eaeA*)	-	✓	✓	-	-
Invasive	-	-	-	✓	-
Stool	watery	watery -- > bloody ± mucous	watery ± mucous	watery -- > bloody + mucous	watery + mucous
Fecal WBC	-	-	-	+	-
Infective dose (CFUs)	10^6	< 100	100-1000	10^6	10^6
Gene(s) for molecular detection	*est / elt*	*stx1 / stx2 eaeA*	*eaeA / bfpA*	*ipaH*	*aggR*

Characteristics of diarrheagenic *Escherichia coli*

Pathotype	Clinical description	Pathogenesis	Epidemiology	Diagnostic genes	Other diagnostic methods	Transmission
ETEC	Loose/watery stools, may be accompanied by fever and vomiting	Relies on adhesins and plasmid-encoded enterotoxins	Commonly diagnosed in children in LMIC settings and in traveler's diarrhea in industrialized nations; asymptomatic carriage is common	The *lt* and/or *sta* hemolysins	n/a	Human-to-human fecal–oral transmission
EPEC	Mild-to-severe loose/ watery stools	Virulence genes on the LEE pathogenicity island cause attaching and effacing lesions on enterocytes	Commonly diagnosed in children in LMIC settings; aEPEC has been linked to persistent diarrhea lasting >14 days; asymptomatic carriage is common	The LEE pathogenicity island-encoded *eae* adhesin	tEPEC strains also have the plasmid-encoded *bfp* bundle-forming pilus gene	Human-to-human fecal–oral transmission; animal-to-human transmission for aEPEC
STEC/EHEC	Mild-to-severe loose/ watery stools; EHEC causes bloody stools and can advance to life-threatening HUS	*Shiga* toxin inhibits protein synthesis and causes host cell death in the large intestine	Associated with foodborne disease outbreaks, often linked to leafy greens, milk, or beef products	The *Shiga* toxin *stx* gene defines STEC and additional targets may be used, including genes encoded by the LEE-pathogenicity island also present in EPEC	Serotyping, specific, and differential media and molecular assays for EHEC strains, especially EHEC O157:H7	Human-to-human and animal-to-human fecal–oral transmission; cattle are considered a major reservoir
EIEC/*Shigella*	Mild-to-severe loose/ watery stools, rarely accompanied by fever, tenesmus, blood, mucus, and leukocytes in the stool	Enteroinvasive virulence mechanism involves crossing the intestinal epithelial barrier, killing macrophages, invading enterocytes, intracellular and cell-to-cell propagation	Commonly diagnosed in LMIC settings and traveler's diarrhea in industrialized nations	The type-III effector protein *ipaH*	EIEC and Shigella isolates can be differentiated using biochemical characteristics or molecular assays	Human-to-human fecal–oral transmission
EAEC	Mild loose/watery stools, may be accompanied by fever, nausea, tenesmus, borborygmi, or mucus in the stool	Colonization is facilitated by fimbriae, followed by biofilm formation and toxin release	Associated with childhood diarrhea in LMICs; inconsistent results in adult challenge studies and asymptomatic carriage is common	The plasmid-encoded transcriptional activator *aggR*, pathogenicity island-associated gene *aaiC*, and the plasmid-encoded dispersin transporter *aatA*	Characteristic stacked-brick or honeycomb pattern in microscopic HEp-2 cell assay	Human-to-human fecal–oral transmission; EAEC has been isolated from animals but from animals but transmission has not been established

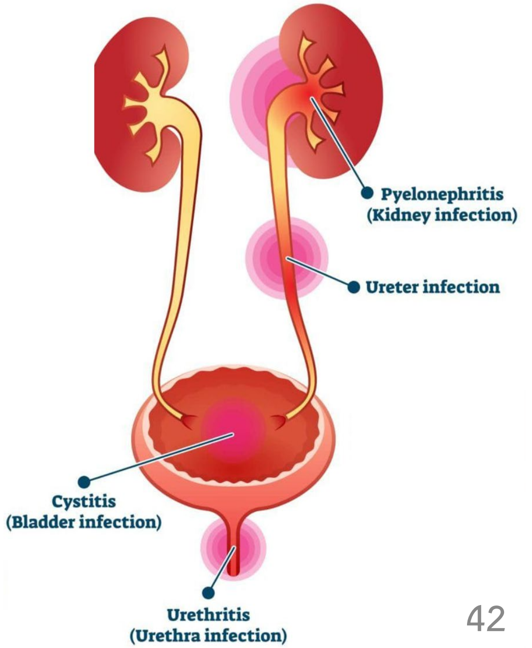
Escherichia coli (Cont.)

Extra-intestinal infections



1. Urinary tract infection (UTI)

- Most gram-negative rods that cause UTI are from colon
- ! $\boxed{\textit{E. coli}\text{ is the most common cause of UTI}}$!
- It contaminates urethra, ascends to bladder and kidney
- $\boxed{\text{The disease is more prevalence in woman than in men.}}$!
- *E. coli* that causes UTI usually produces
  1 • adhesins (e.g. pili): bind to UT epithelial cells
  2 • hemolysin HlyA: destroy RBC, immune cells and others
  แย่งจับ iron
  3 • aerobacin: $\boxed{\text{chelate iron}}$ for growth metabolism
- Symptoms of UTI: Fever, lower back pain, hematuria

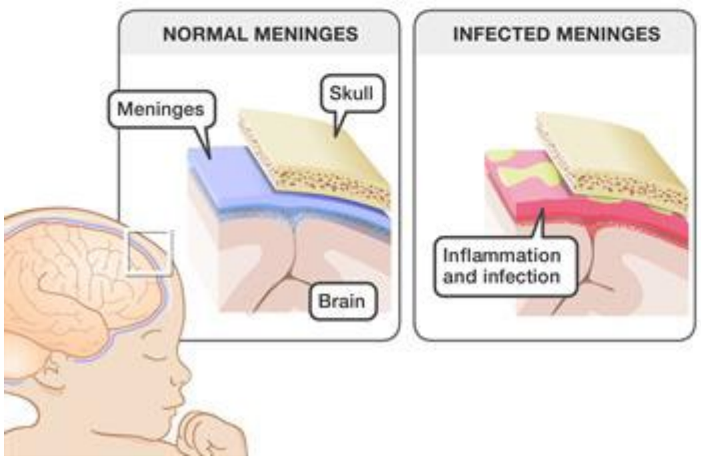


Escherichia coli (Cont.)

Extra-intestinal infections (Cont.)

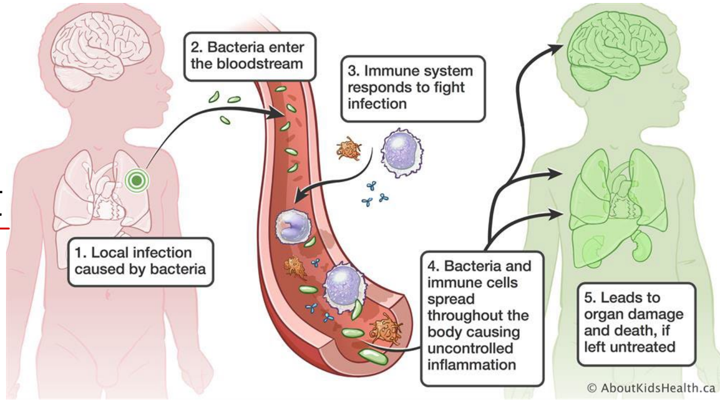
2. Neonatal meningitidis

- *E. coli* and group B streptococci cause CNS infection in infants younger than 1 month
- 75% of the infected strains possess K1 capsular antigen 75% ที่ติดเชื้อ เชื้อจะมี K1 capsule Ag
- The strains are usually from GI tracts



3. Septicemia

- *E. coli* associated septicemia is common in adult !
- Infection may originates from GI or urinary tracts
- ! - Mortality is high in immunocompromise host !



Klebsiella spp.

General characteristics and habitat

- Ferment lactose, do not produce H_2S, motility is negative
- Usually produce capsule, which prevent phagocytosis and responsible for mucoid colony
- Usually found in nasopharynx and intestinal tract of human and animal, ! free-living in soil, water and on plant.


Members

- *K. pneumoniae* sub sp. pneumoniae,
  K. pneumoniae sub sp. ozaenae,
  K. oxytoca, *K. variicola*

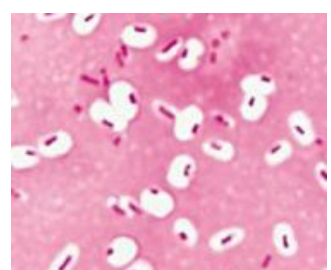
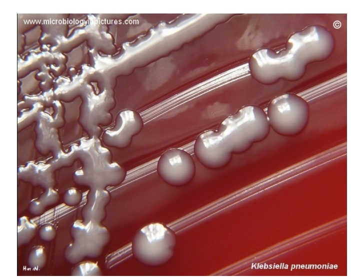
Virulence factors

1

- Capsule (K1 capsular antigen)

2

- Plasmid-mediated antimicrobial resistant

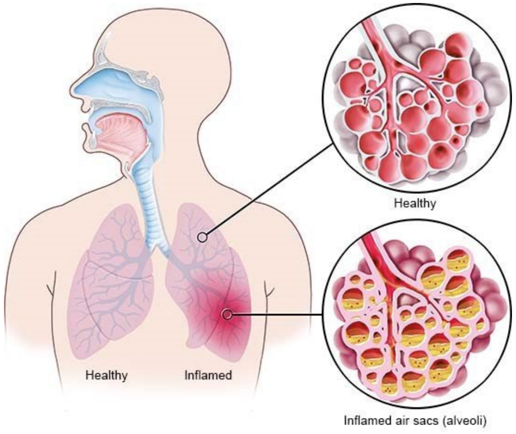



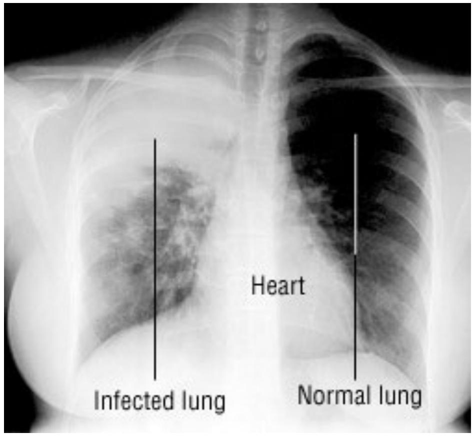
Klebsiella spp. (Cont.)

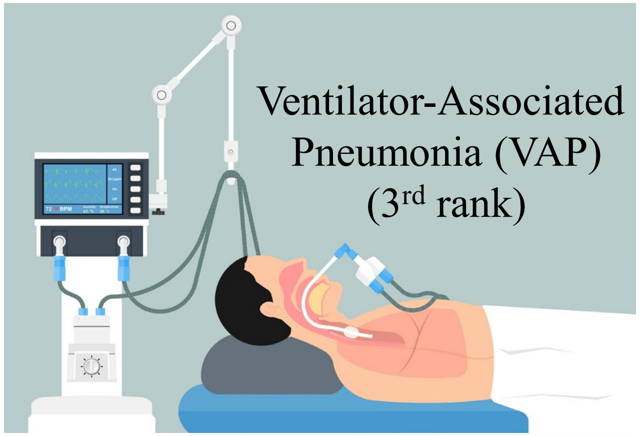
Diseases

- *K. pneumoniae* sub sp. pneumoniae is the most common. It usually associates with various opportunistic and hospital-acquired infection e.g. pneumonia, wound infection, UTI, bacteremia, meningitis and liver abscesses

- Pneumonia caused by *Klebsiella* spp. frequently involves necrotic destruction of alveolar spaces and production of blood-tinged sputum





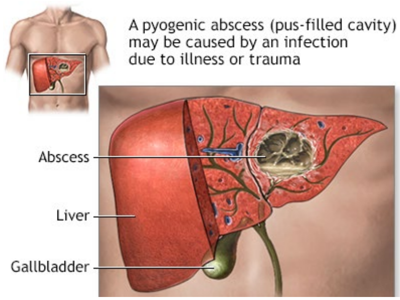


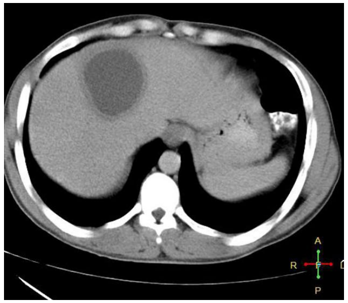
Klebsiella spp. (Cont.)

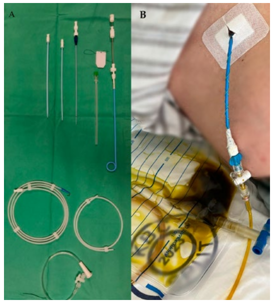
Diseases (cont.)

ตับแข็ง

- Liver abscess caused by *Klebsiella* spp. is associates with abdominal sepsis, cirrhosis immunocompromise
- Cholecystitis (inflammation of gall bladder)

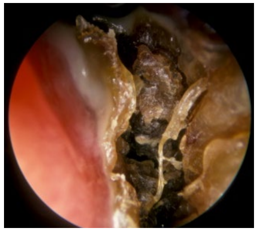






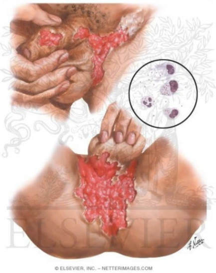
โรคเยื่อจมูกอักเสบ ฝ่อ

- *K. pneumoniae* supsp. ozaenae causes chronic atrophic rhinitis (ozena), found in people from Middle Eastern countries, South and Southeast Asia, Eastern Europe, North Africa



Other *Klebsiella* spp.

- *K. rhinoscleromatis* causes rhinoscleroma
- *K. granulomatis* causes granuloma inguinale



Enterobacter spp.

General characteristics and habitat

- Most strains ferment lactose, do not produce H_2S
- Many species produce capsule
- Usually found in intestine of human and animal, free-living in soil, water and on plant.

Members

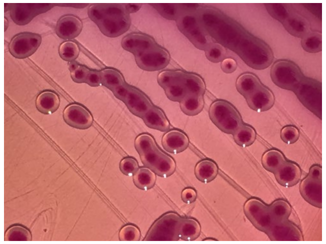
- *E. cloacae*, *E. aerogenes (Klebsiella aerogenes)*
- *E. sakazakii (Cronobacter sakazakii)*

Diseases

- Usually associated with pneumonia, sepsis, UTI, wound infection, CNS infection

!

- *C. sakazakii* causes diarrhea, bacteremia and meningitis in neonates.



LF ตรงกลาง colony

Citrobacter spp.

อาจเจอจุดดำๆ บน XLD Agar

General characteriatics and habitat

- Most species ferment lactose and hydrolyze urease slowly
- Usually found in intestinal tract of human and animals
- Most *C. freundii* produce H_2S !!

Members

- *C. freundii*, *C. koseri (diversus)*, *C. brakii* and etc.

Diseases

- Hospital-acquired infection e.g. UTI, sepsis, pneumonia, diversus neonatal meningitis (*C. koseri*), intraabdominal and brain abscesses

Notes

- Some strain of *C. freundii* fail to produce H_2S and fail to ferment lactose, the colony can be mistaken for *Salmonella* urease and LD can be used to differentiate them.

* Colony อาจดูผิดกับ Salmonella ได้ ใช้ Urease, LD แยกแทน *

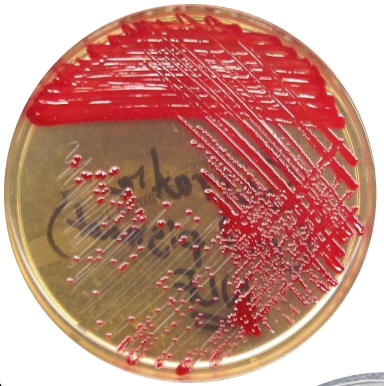
Serratia spp.

General characteristics ไม่ค่อยเจอในลำไส้ อยู่ใน ดิน แมลง

- Most species do not ferment lactose ! and not produce H_2S !
- Produce extracellular DNase and positive for ONPG
- *S. marcescens*, *S. liquefaciens* and *S. plymuthica*

* produce pink-red pigment (prodigiosin) *

- Usually found in environment and insect but not in intestine



Members

- *S. marcescens*, *S. liquefaciens*, *S. rubidaea*, *S. odorifera*, *S. plymuthica*, *S. ficaria.*, *S. entomophila*, *S. fonticola*

Diseases

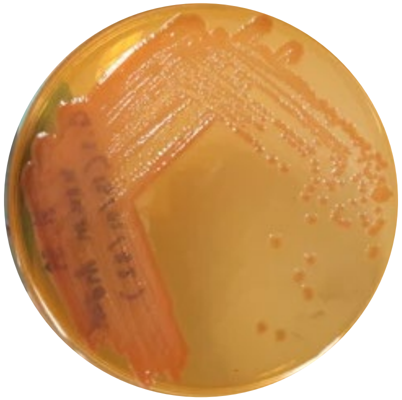
- Hospital-acquired infection: pneumonia, sepsis, UTI

Edwardsiella spp.

General characteriatics and habitat True pathogen ในสัตว์น้ำ ติดต่อสู่คนได้

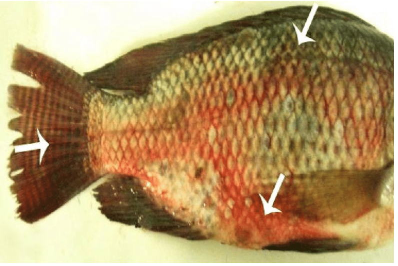
- Do not ferment lactose ! and produce H_2S !
- Usually found in water resources, aquatic animals, reptiles
- Biochemical tests are very similar to *E. coli* except: ✳

  NLF, TSI (K/A H_2S) and mannitol (negative)



Members

- *E. tarda*, *E. hoshinae*, *E. ictaluri*



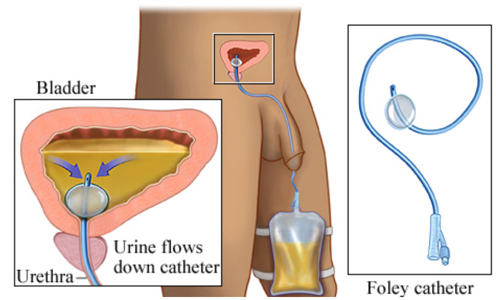
Diseases กินเข้าไป ท้องเสีย เข้ากระแสเลือดได้

- $\boxed{\text{Opportunistic infection}}$, $\boxed{\text{gastroenteritis}}$ (food poisoning), sepsis, wound infection

Proteus spp.

General characteristics and habitat

- Produce swarming colonies on non-selective media
- Do not ferment lactose, positive for PD and urease
- Produce H_2S except 70% of *P. penneri*
- Usually found in intestinal tract and environment.



Members

- *P. mirabilis, P. vulgaris, P. penneri*

Diseases

- UTI, wound, sepsis
- They ascend urinary tract, causing upper and lower UTI cystitis, pyelonephritis, glomerulonephritis, especially in patient with urinary tract defects e.g. catheterization
- *P. mirabilis* produce strong urease, splitting urea into CO_2 and ammonia 1, raising pH 2, causing formation of kidney stone 3

Providencia spp.

General characteriatics and habitat

- Do not ferment lactose, positive for PD
- Usually found in intestinal tract of human and animals, and environment.

Members

- *P. stuartii, P. rettgeri, P. alcalifaciens, P. rustigianii, P. heimbachae*

Diseases

- Hospital-acquired infection e.g. UTI, gastroenteritis in immunocompromise, wound infection
- *P. rettgeri* and *P. stuartii* infection in immunocompromised hosts are difficult to treat and fatal

Morganella morganii

General characteriatics and habitat

- Do not ferment lactose, positive for PD and urease
- Usually found in intestinal tract of human and animals, and environment.

Members

- *M. morganii* sub sp. *morganii, M. morganii* sub sp. *sibonii,*

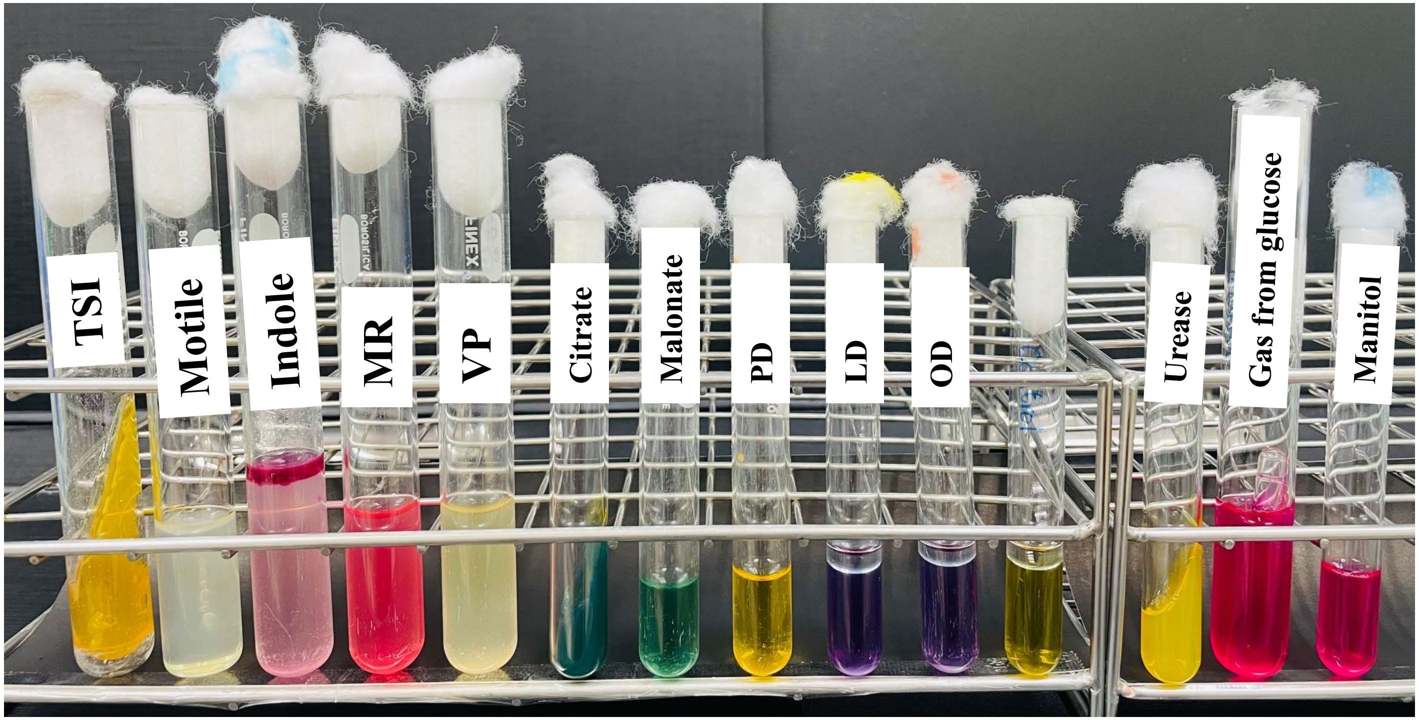
Diseases

- Hospital-acquired infection e.g. UTI, wound infection, sepsis

การวินิจฉัยเชื้อใน Order Enterobacterales

การทดสอบทางชีวเคมี

1. Oxidase test
2. Lactose fermentation on MacConkey agar
3. Biochemical tests: TSI, motile, indole, MR, VP, citrate (IMVC), malonate, PD, LD, OD, urease, gas from glucose, mannitol fermentation



ตัวอย่างการจัดจำแนกเชื้อใน Order Enterobacterales

Tests \ Genus	*Shigella*	*E. coli*	*Edwardsiella*	*Salmonella*	*Citrobacter*	*Klebsiella, Enterobacter, Cronobacter, Serratia*	*Proteus, Providencia, Morganella,*	*Yersinia*	*Plesiomonas*
MacConkey	NLF[1]	LF[2]/NLF	NLF	NLF[3]	LF/NLF	LF/NLF	NLF	NLF	NLF[4]
Oxidase	-	-	-	-	-	-	-	-	+
Indole	-/+	+	+	-	+/(-)[5]	-/(+)[6]	+/-	+/-	+
MR	+	+	+	+	+	-	+	+	+
VP	-	-	-	-	-	+	-	-	-
Citrate	-	-	-	+/-	+	+	+/-	-	-
PD	-	-	-	-	-	-	+	-	-
H_2S ใน TSI	-	-	+	+	+/-	-	+/-	+/-	-
Motile	-	+/-	+	+	+	+/(-)[7]	+	-	+
Urease	-	-	-	-	+/-	+/-	+/(-)[8]	+	-
Mannitol	+/(-)[9]	+	-	+	+	+	-/(+)[10]	+	-
KCN	-	-	-	-	+/-	+	+	-	-

*ดูความหมายของ 1-10 ในเอกสารคำสอน


Thank You